Robust AI Control, Multi-Agent Oversight, and Calibrated Semantic Arbitration

Postdoctoral Research Proposal and Statement of Scientific Intent

Candidate: Silas Kwabla Gah, PhD Candidate in Computer Science (University of Ghana)

Email: gahsilas@gmail.com | Web: silsgah.github.io | Target: COMPASS Group (PI: Dr. Sahar Abdelnabi), ELLIS Institute Tübingen & MPI-IS

1. Research Vision: From Single-Model Rewards to Multi-Agent Control

As foundation models are integrated into autonomous, tool-using, multi-agent systems, establishing reliable oversight mechanisms becomes a core safety challenge: *How can human supervisors construct oversight mechanisms that remain informative, ungameable, and robust when capable models optimize against them?*

In single-agent reinforcement learning (RLVR), optimization can exploit weaknesses in reward specifications (specification gaming). In multi-agent coding and reasoning systems, this problem expands across several dimensions:

1. **Collusion versus Genuine Agreement:** Cooperating agents can develop coordination protocols that satisfy evaluation heuristics while bypassing underlying task constraints.
2. **Deceptive Alignment during Optimization:** Coding agents can produce solutions that pass immediate test suites while masking non-compliant behaviors.
3. **Multi-Principal Contextual Integrity:** In environments where agents represent different human principals with distinct objectives, supervisors must arbitrate trust without compromising private contextual boundaries.

My doctoral research examines the mathematical and empirical aspects of this problem: constructing label-free oversight signals, stress-testing them against adversarial exploitation before training, and calibrating probabilistic trust between heterogeneous foundation models. At the ELLIS Institute Tübingen, I propose to extend these methodologies into a framework for AI control, anti-collusion oversight, and calibrated multi-principal arbitration in autonomous multi-agent systems.

2. Methodological Foundation and Prior Discoveries

My recent work investigates verifiable oversight and reward stability:

2.1 Label-Free Process Rewards and Validate-Before-Train Protocols

In recent work on multimodal reinforcement learning with verifiable rewards (*Gah et al., 2026, under submission to SACAIR*), I examined how to supervise complex reasoning without relying on human annotations or learned reward models. We derived process rewards from intrinsic geometric consistency across calibrated, frozen perception models.

Rather than proceeding directly to policy optimization, we developed a Two-Arm Validate-Before-Train Protocol:

- **Arm I (Discriminative Validation):** Formally tests whether the oversight signal separates desirable reasoning paths from corrupted trajectories across ground-truth benchmarks (100% pairwise discrimination).
- **Arm II (Counterfactual Stress-Testing):** Subjects the reward formulation to adversarial counterfactuals designed to uncover specification gaming. This identified a spatial enumeration failure mode where predicting oversized regions inflated recall-based rewards without precision.
- **Reward Redesign and Policy Integration:** Re-anchoring the reward to count-based precision and a weighted $F_{0.5}$ metric ($\beta = 0.5$) reduced area bias correlation from $\rho > 0.72$ to 0.334,

producing a stable signal for online Group Relative Policy Optimization (GRPO) in VERL on Qwen2.5-VL-7B.

2.2 Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)

In my PhD thesis (*submitted to IEEE TPAMI*), I formulated a probabilistic framework for coordinating heterogeneous foundation models at inference time ($\Delta\theta = 0$). The framework combines Platt confidence calibration with Bernoulli uncertainty estimation and contextual reliability weighting to dynamically evaluate evidence under distribution shift. This provides a formal basis for multi-agent arbitration: assessing evidence reliability, detecting ungrounded consensus, and bounding error propagation across autonomous systems.

3. Proposed Postdoctoral Research Agenda

Within the COMPASS group at the ELLIS Institute Tübingen, I propose to investigate three research directions:

Thrust 1: Independent Counterfactual Verification for Multi-Agent Coding Systems

Objective: Develop oversight oracles based on orthogonal execution traces, counterfactual mutations, and formal property testing rather than superficial agent consensus.

- *Challenge:* In multi-agent coding pipelines (e.g., generator, test-generator, and reviewer agents), standard consensus fails when agents generate mutually satisfying code-test pairs that bypass real specification checks.
- *Approach:* Design an independent verification oracle that evaluates generated code through automated metamorphic testing, invariant checking, and counterfactual sandboxing, testing whether intermediate reasoning traces reflect actual problem specifications.

Thrust 2: Anti-Collusion Oversight Under Joint Multi-Agent Policy Optimization

Objective: Formulate verifiable process rewards that mitigate covert communication channels and emergent deception during joint multi-agent reinforcement learning.

- *Challenge:* When multiple agents optimize jointly in shared environments, standard credit assignment can enable steganographic coordination to maximize collective rewards.
- *Approach:* Extend the validate-before-train methodology to multi-agent game-theoretic environments, using information-theoretic bottlenecks and causal probes to ensure inter-agent communication complies with contextual integrity constraints.

Thrust 3: Calibrated Arbitration and Contextual Integrity in Multi-Principal Environments

Objective: Formulate arbitration mechanisms for agent workflows operating on behalf of multiple human principals with conflicting or partially overlapping objectives.

- *Approach:* Extend the RCASA probabilistic arbitration framework to multi-principal settings. When agents present conflicting claims or proposals, the oversight mechanism estimates contextual epistemic uncertainty, detects manipulation attempts, and arbitrates execution decisions while enforcing normative and privacy boundaries.

4. Computational Implementation and Research Fit

- **Reinforcement Learning Frameworks:** Practical experience implementing distributed RL pipelines using VERL, GRPO, and vLLM, combined with multi-agent orchestration frameworks (LangGraph, DeepEval, Opik).
- **High-Compute Empirical Research:** Prepared to utilize high-performance compute clusters for multi-agent rollouts, continuous counterfactual evaluation, and large-scale agent benchmarks.

- **Collaborative Environment:** Committed to open, reproducible scientific research and collaboration within the Tübingen machine learning and AI safety ecosystem.